

Machine Learning I — Hands-On 5 — Worked Solutions

5.1 Linear regression — worked solution

1. Parameter estimation

The first dataset is

Example	x	y
1	1	2
2	2	5
3	4	4
4	6	8
5	7	7

We fit

$$\hat{y} = w_0 + w_1 x,$$

with

$$w_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad w_0 = \bar{y} - w_1 \bar{x}.$$

Step 1 — Compute the means

$$\bar{x} = \frac{1 + 2 + 4 + 6 + 7}{5} = \frac{20}{5} = 4.00,$$

$$\bar{y} = \frac{2 + 5 + 4 + 8 + 7}{5} = \frac{26}{5} = 5.20.$$

Step 2 — Compute the numerator of w_1

x_i	y_i	$x_i - 4.00$	$y_i - 5.20$	$(x_i - \bar{x})(y_i - \bar{y})$
1	2	-3.00	-3.20	9.60
2	5	-2.00	-0.20	0.40
4	4	0.00	-1.20	0.00
6	8	2.00	2.80	5.60
7	7	3.00	1.80	5.40

Therefore

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = 9.60 + 0.40 + 0.00 + 5.60 + 5.40 = 21.00.$$

Step 3 — Compute the denominator of w_1

x_i	$x_i - 4.00$	$(x_i - \bar{x})^2$
1	-3.00	9.00
2	-2.00	4.00
4	0.00	0.00
6	2.00	4.00
7	3.00	9.00

Hence

$$\sum_i (x_i - \bar{x})^2 = 9 + 4 + 0 + 4 + 9 = 26.00.$$

Step 4 — Compute w_1

$$w_1 = \frac{21.00}{26.00} \approx 0.81.$$

Step 5 — Compute w_0

$$\begin{aligned} w_0 &= \bar{y} - w_1 \bar{x} \\ &= 5.20 - (0.81)(4.00) \\ &= 5.20 - 3.24 \\ &= 1.96. \end{aligned}$$

Final regression equation

$$\boxed{\hat{y} = 1.96 + 0.81x}.$$

2. Prediction for $x = 6$

Using the fitted model,

$$\begin{aligned} \hat{y}(6) &= 1.96 + 0.81(6) \\ &= 1.96 + 4.86 \\ &= 6.82. \end{aligned}$$

Thus

$$\boxed{\hat{y}(6) = 6.82}.$$

3. Mean Squared Error

Step 1 — Compute all predictions

x_i	y_i	$\hat{y}_i$
1	2	$1.96 + 0.81(1) = 2.77$
2	5	$1.96 + 0.81(2) = 3.58$
4	4	$1.96 + 0.81(4) = 5.20$
6	8	$1.96 + 0.81(6) = 6.82$
7	7	$1.96 + 0.81(7) = 7.63$

Step 2 — Compute residuals and squared residuals

x_i	y_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$	e_i^2
1	2	2.77	-0.77	0.59
2	5	3.58	1.42	2.02
4	4	5.20	-1.20	1.44
6	8	6.82	1.18	1.39
7	7	7.63	-0.63	0.40

The total squared error is

$$0.59 + 2.02 + 1.44 + 1.39 + 0.40 = 5.84.$$

Step 3 — Compute the final MSE

$$\text{MSE} = \frac{5.84}{5} = 1.168 \approx 1.17.$$

Therefore

$$\boxed{\text{MSE} \approx 1.17}.$$

5.2 Dataset (a): fit the linear model

Dataset (a) contains

Example	x	y
1	1	3
2	2	4
3	3	2
4	4	6
5	5	5
6	6	8
7	7	7
8	8	10
9	9	9
10	10	13

Step 1 — Compute the means

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10}{10} = \frac{55}{10} = 5.50,$$

$$\bar{y} = \frac{3 + 4 + 2 + 6 + 5 + 8 + 7 + 10 + 9 + 13}{10} = \frac{67}{10} = 6.70.$$

Step 2 — Compute the numerator of w_1

x_i	y_i	$x_i - 5.50$	$y_i - 6.70$	Product
1	3	-4.50	-3.70	16.65
2	4	-3.50	-2.70	9.45

x_i	y_i	$x_i - 5.50$	$y_i - 6.70$	Product
3	2	-2.50	-4.70	11.75
4	6	-1.50	-0.70	1.05
5	5	-0.50	-1.70	0.85
6	8	0.50	1.30	0.65
7	7	1.50	0.30	0.45
8	10	2.50	3.30	8.25
9	9	3.50	2.30	8.05
10	13	4.50	6.30	28.35

Thus

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = 85.50.$$

Step 3 — Compute the denominator

x_i	$x_i - 5.50$	$(x_i - \bar{x})^2$
1	-4.50	20.25
2	-3.50	12.25
3	-2.50	6.25
4	-1.50	2.25
5	-0.50	0.25
6	0.50	0.25
7	1.50	2.25
8	2.50	6.25
9	3.50	12.25
10	4.50	20.25

Hence

$$\sum_i (x_i - \bar{x})^2 = 82.50.$$

Step 4 — Compute w_1

$$w_1^{(a)} = \frac{85.50}{82.50} \approx 1.04.$$

Step 5 — Compute w_0

$$\begin{aligned} w_0^{(a)} &= 6.70 - (1.04)(5.50) \\ &= 6.70 - 5.72 \\ &= 0.98. \end{aligned}$$

The fitted model is

$$\boxed{\hat{y}^{(a)} = 0.98 + 1.04x}.$$

5.3 Dataset (a): predictions and residuals

Using $\hat{y}^{(a)} = 0.98 + 1.04x$:

Ex.	x	y	$\hat{y}^{(a)}$	$e_i = y_i - \hat{y}_i^{(a)}$
1	1	3	2.02	0.98
2	2	4	3.06	0.94
3	3	2	4.10	-2.10
4	4	6	5.14	0.86
5	5	5	6.18	-1.18
6	6	8	7.22	0.78
7	7	7	8.26	-1.26
8	8	10	9.30	0.70
9	9	9	10.34	-1.34
10	10	13	11.38	1.62

The largest residual in absolute value is Example 3:

$$|-2.10| = 2.10.$$

The residual signs are reasonably mixed, with no single observation strongly pulling the entire fit in one direction.

5.4 Dataset (b): fit the linear model

Dataset (b) is identical except that the final target is changed from 13 to 20.

Step 1 — Compute the means

The inputs have not changed, so

$$\bar{x} = 5.50.$$

The new target mean is

$$\bar{y} = \frac{3 + 4 + 2 + 6 + 5 + 8 + 7 + 10 + 9 + 20}{10} = \frac{74}{10} = 7.40.$$

Step 2 — Compute the numerator of w_1

x_i	y_i	$x_i - 5.50$	$y_i - 7.40$	Product
1	3	-4.50	-4.40	19.80
2	4	-3.50	-3.40	11.90
3	2	-2.50	-5.40	13.50
4	6	-1.50	-1.40	2.10
5	5	-0.50	-2.40	1.20
6	8	0.50	0.60	0.30
7	7	1.50	-0.40	-0.60
8	10	2.50	2.60	6.50
9	9	3.50	1.60	5.60

x_i	y_i	$x_i - 5.50$	$y_i - 7.40$	Product
10	20	4.50	12.60	56.70

Therefore

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = 117.00.$$

The last observation contributes 56.70 to this sum by itself, which already shows how strongly it affects the fitted slope.

Step 3 — Compute the denominator

The x values are unchanged, so

$$\sum_i (x_i - \bar{x})^2 = 82.50.$$

Step 4 — Compute w_1

$$w_1^{(b)} = \frac{117.00}{82.50} \approx 1.42.$$

Step 5 — Compute w_0

$$\begin{aligned} w_0^{(b)} &= 7.40 - (1.42)(5.50) \\ &= 7.40 - 7.81 \\ &= -0.41. \end{aligned}$$

The fitted model is

$$\hat{y}^{(b)} = -0.41 + 1.42x.$$

5.5 Dataset (b): predictions and residuals

Ex.	x	y	$\hat{y}^{(b)}$	$e_i = y_i - \hat{y}_i^{(b)}$
1	1	3	1.01	1.99
2	2	4	2.43	1.57
3	3	2	3.85	-1.85
4	4	6	5.27	0.73
5	5	5	6.69	-1.69
6	6	8	8.11	-0.11
7	7	7	9.53	-2.53
8	8	10	10.95	-0.95
9	9	9	12.37	-3.37
10	10	20	13.79	6.21

The largest absolute residual is now Example 10:

$$|6.21| = 6.21.$$

The outlier has pulled the line upward on the right. As a result, several of the remaining high- x observations now have negative residuals. The residual pattern is therefore more structured than in dataset (a).

5.6 Compare dataset (a) and dataset (b)

The two fitted models are

$$\hat{y}^{(a)} = 0.98 + 1.04x,$$

and

$$\hat{y}^{(b)} = -0.41 + 1.42x.$$

Step 1 — How much did the slope change?

$$1.42 - 1.04 = 0.38.$$

The slope increased by approximately 0.38.

Step 2 — Did the regression line shift substantially?

Yes. The intercept changed from approximately 0.98 to -0.41 , while the slope increased from approximately 1.04 to 1.42. The second line is therefore steeper and is pulled upward toward the extreme observation at large x .

Step 3 — Compare predictions at the same input values

x	$\hat{y}^{(a)}$	$\hat{y}^{(b)}$
1	2.02	1.01
2	3.06	2.43
3	4.10	3.85
4	5.14	5.27
5	6.18	6.69
6	7.22	8.11
7	8.26	9.53
8	9.30	10.95
9	10.34	12.37
10	11.38	13.79

For small x , the predictions are relatively close. As x increases, the predictions from dataset (b) become increasingly larger because its fitted slope is steeper.

Changing a single observation therefore affects **the entire fitted model**, not just the prediction near that point.

Final conclusion

Ordinary least squares is highly sensitive to outliers because

- it is a **global model**, so every observation contributes to the final parameters;
- it minimizes **squared error**, so large deviations have disproportionately large influence;
- changing one extreme observation can alter both slope and intercept and therefore modify predictions across the full input range.

In this example, changing only the final point from $(10, 13)$ to $(10, 20)$ increases the slope from approximately 1.04 to 1.42 and changes the intercept from approximately 0.98 to -0.41 .

i ML insight — influence is a property of the fitting criterion

The outlier is highly impactful because of the geometry of ordinary least squares: the model is global and its objective squares residuals. Those two properties explain the change in fitted parameters.